Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

4 solution use cases · 12 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

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Multi-cloud & Terraform

Hybrid and multi-cloud provisioning, Terraform integration and cloud governance.

Solution use cases

Use Case: Run Terraform from Automation Platform

Description	Organizations that already describe infrastructure in Terraform can run it from Automation Platform and keep one place for credentials, approvals and audit. Terraform creates the cloud resources, and Ansible configures what Terraform built, in a single workflow.
Collection	`cloud.terraform` (Certified), `cloud.terraform_ops` (Validated)
Key content	• `cloud.terraform.terraform` — Manages a Terraform deployment (and plans) • `cloud.terraform.terraform_output` — Returns Terraform module outputs. • `cloud.terraform.plan_stash` — Handle the base64 encoding or decoding of a terraform plan file • `cloud.terraform.git_plan` — Clone a Git repository and apply a plan from it. • `cloud.terraform.inventory_from_outputs` — Create an in-memory inventory from Terraform outputs. • `cloud.terraform.terraform_state` — Builds an inventory from resources created by cloud providers.
Notes	• `cloud.terraform_ops` provides validated roles and playbooks for this integration with Automation Platform; check the collection documentation for the current content. • For HCP Terraform or Terraform Enterprise, see Automate HCP Terraform workspaces and policy checks. • Keep Terraform state out of Automation Platform, and use a remote backend with locking. • Compare with native modules in Provision AWS networking and compute.

Use Case: Provision consistently across clouds

Description	Offer the same service, for example a hardened application server, on more than one public cloud without each cloud team writing its own process. A single request and workflow provisions on the chosen cloud, then applies the same configuration, so the result is identical regardless of provider.
Collection	`cloud.terraform` (Certified), `amazon.aws` (Certified), `azure.azcollection` (Certified), `google.cloud` (Certified)
Key content	• `cloud.terraform.terraform` — Manages a Terraform deployment (and plans) • `cloud.terraform.terraform_provider` — Builds an inventory from Terraform state file. • `cloud.terraform.tf_output` — Reads state file outputs. • `amazon.aws.aws_ec2` — EC2 inventory source • `azure.azcollection.azure_rm` — Azure Resource Manager inventory plugin • `google.cloud.gcp_compute` — Google Cloud Compute Engine inventory source

Notes	• Keep the cloud-specific layer thin; differences in networking and identity are best handled in each provisioning job. • Related: Provision AWS networking and compute, Provision Azure networking and virtual machines and Provision Google Cloud networking and compute. • Enforce the tagging standard with Enforce tagging and cost hygiene across clouds.

Use Case: Enforce tagging and cost hygiene across clouds

Description	Cost reports and ownership depend on consistent tags. Run a recurring job in every cloud that finds resources missing required tags, applies tags or notifies owners, and removes temporary resources once they have expired.
Collection	`amazon.aws` (Certified), `cloud.aws_ops` (Validated), `azure.azcollection` (Certified), `hashicorp.terraform` (Certified)
Key content	• `amazon.aws.ec2_instance_info` — Gather information about ec2 instances in AWS • `amazon.aws.ec2_tag` — Create and remove tags on ec2 resources • `cloud.aws_ops.ec2_instance_terminate_by_tag` — A role to terminate the ec2_instances having a specific tag. • `azure.azcollection.azure_rm_tags` — Manage tags • `hashicorp.terraform.reserved_tag_keys` — Manage Terraform Cloud/Enterprise reserved tag keys (create, update, delete).
Notes	• Start in report-only mode, and enable removal only for resources with an explicit expiry tag. • Google Cloud uses labels instead of tags; apply them through the resource modules in `google.cloud`. • Pair with Event-Driven Ansible to tag new resources when they are created. • Related: Provision consistently across clouds.

Use Case: Automate HCP Terraform workspaces and policy checks

Description	For teams that use HCP Terraform or Terraform Enterprise, Ansible can manage the platform itself: projects, workspaces, variables and policies, and can start runs and decide whether to apply them. Application teams get a ready workspace on request, and governance rules are enforced consistently.
Collection	`hashicorp.terraform` (Certified)

Key content	• `hashicorp.terraform.workspace` — Manage workspaces in Terraform Enterprise/Cloud. • `hashicorp.terraform.variable_sets` — Manage Terraform Cloud/Enterprise variable sets (create, update, delete, attach). • `hashicorp.terraform.run` — Manage Terraform Cloud/Enterprise runs (create, apply, cancel, discard). • `hashicorp.terraform.policy_set` — Manage Terraform Cloud/Enterprise policy sets and their memberships. • `hashicorp.terraform.plan_analyze` — Analyze a Terraform plan for drift and change classification • `hashicorp.terraform.promote_run` — Gate and apply a Terraform Cloud/Enterprise run based on policy outcomes. • `hashicorp.terraform.tfc_inv` — Unified dynamic inventory plugin for HCP Terraform / Terraform Enterprise.
Notes	• Combine with Run Terraform from Automation Platform if some configurations still run with the Terraform command line. • Treat API tokens as secrets, and scope them to the organization or team that needs them. • Pair with Event-Driven Ansible to start a run when a webhook reports a change in the source repository.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Ansible turbo mode `cloud.common`

Certified Red Hat · version 5.0.0 · updated 2025-06-25 · 3 modules → 1 use cases

Use Case: Test and demonstrate turbo mode

Description	Provide demo, failure and import modules and a lookup for developing and testing turbo mode.
Collection	`cloud.common` (Certified, Red Hat)
Key content	**Demo module** `turbo_demo` **Failure module** `turbo_fail` **Import test module** `turbo_import` **Demo lookup** lookup `turbo_demo`

HashiCorp Terraform `cloud.terraform`

Certified Red Hat · version 4.0.0 · updated 2025-07-11 · 3 modules → 2 use cases

Use Case: Run Terraform deployments and plans

Description	Apply Terraform configurations and plans, read module outputs, stash plan files and apply plans from a Git repository.
Collection	`cloud.terraform` (Certified, Red Hat)
Key content	**Terraform deployments** `terraform` **Terraform outputs** `terraform_output` **Plan stash** `plan_stash` **Apply a plan from Git** role `git_plan` **Terraform output lookup** lookup `tf_output`

Use Case: Build inventory from Terraform state

Description	Build Ansible inventory from Terraform state files and outputs.
Collection	`cloud.terraform` (Certified, Red Hat)
Key content	**Terraform state inventory (providers)** inventory `terraform_provider` **Terraform state inventory (cloud resources)** inventory `terraform_state` **In-memory inventory from outputs** role `inventory_from_outputs`

HashiCorp Terraform (HCP Terraform and Terraform Enterprise)

`hashicorp.terraform`

Certified HashiCorp | An IBM Company · version 2.2.1 · updated 2026-09-04 · 87 modules → 9 use cases

Use Case: Manage organizations, projects and workspaces

Description	Create and maintain HCP Terraform and Terraform Enterprise organizations, projects and workspaces, with workspace bootstrap and configuration versions.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Organizations** `organizations` **Projects** `project`, `project_info` **Workspaces** `workspace`, `workspace_info` **Workspace bootstrap** `workspace_bootstrap` **Configuration versions** `configuration_version`, `configuration_version_info` **Organization tags** `organization_tags` **Reserved tag keys** `reserved_tag_keys` **Explorer saved views** `explorer`, `explorer_info` **No-code modules** `no_code_module`, `no_code_module_info`

Use Case: Run plans and applies

Description	Create, apply, cancel and discard runs, view plans, analyze plans for drift, and read run outputs and state outputs.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Runs** `run`, `run_info` **Policy-gated run promotion** `promote_run` **Run triggers** `run_trigger` **Plan details** `view_plan` **Plan drift analysis** `plan_analyze` **State version outputs** `output`

Use Case: Manage variables and variable sets

Description	Manage workspace variables and variable sets and attach them to workspaces and projects.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Variables** `variable` **Variable sets** `variable_sets`

Use Case: Enforce policy and run tasks

Description	Manage Sentinel, OPA and tf-policy policies and policy sets, review evaluations and task results, override checks, and attach run tasks.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Policies** `policy`, `policy_info` **Policy sets** `policy_set`, `policy_set_info` **Policy set parameters** `policy_set_parameter`, `policy_set_parameter_info` **Policy set versions** `policy_set_version`, `policy_set_version_info` **Policy check overrides** `policy_check`, `policy_check_info` **OPA policy evaluations** `policy_evaluation_info` **Policy set outcomes** `policy_set_outcome_info` **tf-policy evaluation overrides** `tf_policy_evaluation`, `tf_policy_evaluation_info` **Run tasks** `run_task`, `run_task_info` **Workspace run tasks** `workspace_run_task`, `workspace_run_task_info` **Task stages** `task_stage_info` **Task results** `task_result_info` **Notification configurations** `notification_configuration`

Use Case: Control access with teams and tokens

Description	Manage teams, their access to projects and workspaces, user lookups, and organization, team and agent tokens.

Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Teams** `team`, `team_info` **Team project access** `team_project_access` **Team workspace access** `team_workspace_access` **Users** `user_info` **Organization tokens** `organization_token` **Team tokens** `team_token`, `team_token_info` **SSH keys** `ssh_keys`

Use Case: Run agents and manage dynamic credentials

Description	Manage self-hosted agent pools, agents and tokens, and OIDC and HYOK (Hold Your Own Key) configurations for AWS, Azure, GCP and Vault.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Agent pools** `agent_pool`, `agent_pool_info` **Agents** `agent`, `agent_info` **Agent tokens** `agent_token`, `agent_token_info` **HYOK configurations** `hyok_configuration`, `hyok_configuration_info` **AWS OIDC configurations** `aws_oidc_configuration`, `aws_oidc_configuration_info` **Azure OIDC configurations** `azure_oidc_configuration`, `azure_oidc_configuration_info` **GCP OIDC configurations** `gcp_oidc_configuration`, `gcp_oidc_configuration_info` **Vault OIDC configurations** `vault_oidc_configuration`, `vault_oidc_configuration_info`

Use Case: Publish and consume registry content

Description	Manage private registry modules and providers with their versions and platforms, and look up public registry modules.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Private registry modules** `registry_module`, `registry_module_info` **Private registry providers** `registry_provider`, `registry_provider_info` **Provider versions** `registry_provider_version`, `registry_provider_version_info` **Provider platforms** `registry_provider_platform`, `registry_provider_platform_info` **Public registry modules** `public_registry_module_info`

Use Case: Manage Terraform stacks

Description	Create stacks and configuration snapshots, and inspect deployment groups, runs, steps, diagnostics and state.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)

Key content	**Stacks** `stack`, `stack_info` **Stack configurations** `stack_configuration`, `stack_configuration_info` **Deployment groups** `stack_deployment_group_info` **Deployment runs** `stack_deployment_run_info` **Deployment steps** `stack_deployment_step_info` **Diagnostics** `stack_diagnostic_info` **Stack state** `stack_state_info`

Use Case: Use inventory, lookups and plan-safety checks in playbooks

Description	Build dynamic inventory from HCP Terraform workspaces, look up outputs, tags, variables, policy checks and run events, and gate automation on drift rules.
Collection	`hashicorp.terraform` (Certified, HashiCorp | An IBM Company)
Key content	**Dynamic inventory** inventory `tfc_inv` **Output values** lookup `tf_output` **Organization tags** lookup `tf_org_tags` **Variable set variables** lookup `tf_variable_set_vars` **Policy check outcomes** lookup `tf_policy_checks` **Run events** lookup `tf_run_events` **Plan guard filter** filter `plan_guard` **Plan safe test** test `plan_safe`

Other certified and validated collections

Module-level content for these collections could not be read from a public source repository. Browse them on Automation Hub.

`cloud.terraform_ops`	A set of Roles and Playbooks used to create/configure Cloud providers' resources as part of AAP Terraform integration. (Validated, Red Hat)